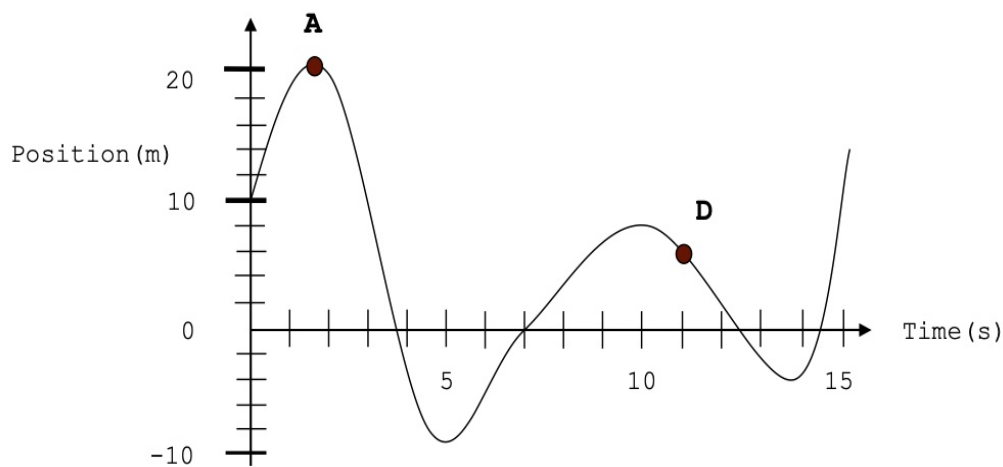
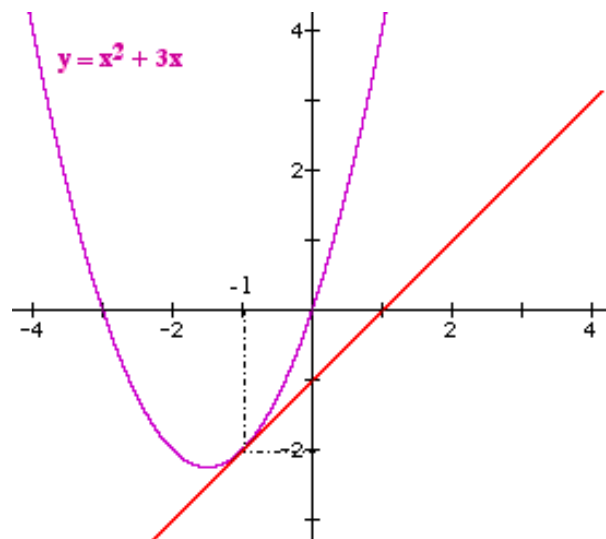


Assignment-02

1. Consider the function $f(x) = 2 - x^2$; $x_0 = -1$ and $x_1 = 2$.
 - (a). Find the instantaneous rate change of function at x_0 .
 - (b). Find the average rate of change of function in the interval $[x_0, x_1]$.
 - (c). Find the equation of the tangent line and secant line along with the graph of $f(x)$.
 - (d). Explain your findings in (c).
2. The following figure represents a position function of a particle at time t .
 - (a). Find the average velocity over the time from A to D.
 - (b). Find the value(s) of t at which the instantaneous velocity is zero.
 - (c). Roughly sketch the velocity graph of the particle.



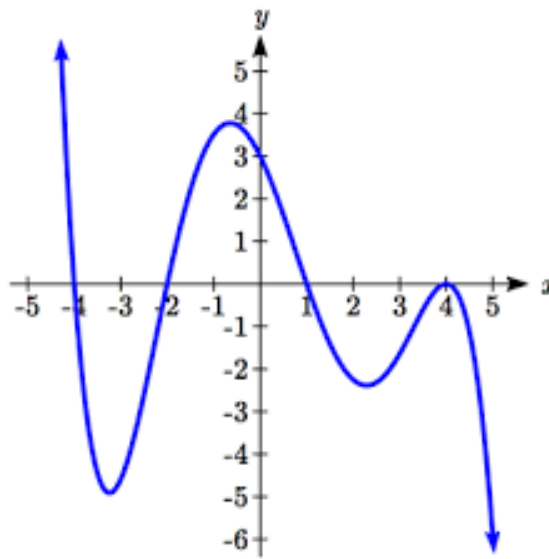
3. Using the definition find the slope of the following curve at the prescribed point drawn below and hence find the equation of the given line.



4. Using definition find the derivative with respect to x of $f(x) = \frac{1}{2x}$, and use it to find the equation of the tangent line to $f(x)$ at $x = 3$.

Assignment-02

5. Check the continuity and differentiability of the function $f(x) = \begin{cases} x^2 - 4 & ; x \leq 2 \\ -x + 2 & ; x > 2 \end{cases}$ at $x = 2$. Also sketch the graph of function.
6. Test the continuity and differentiability of $f(x) = \begin{cases} -x & ; x < 0 \\ x^2 & ; x \geq 0 \end{cases}$ at $x = 0$? Justify your answer graphically.
7. Sketch the graph of $f(x) = \begin{cases} x^2 - 1 & , x \leq 2 \\ 2x - 1 & , x > 2 \end{cases}$. Determine whether the function $f(x)$ is continuous and differentiable at $x = 2$.
8. Sketch the derivative graph of the function given in the following figure.



9. The following table defines the values of $f(x)$, $g(x)$, $f'(x)$ and $g'(x)$ at x . For $u(x) = f(x)g(x)$, and $v(x) = \frac{f(x)}{g(x)}$, find $u'(-1)$, and $v'(0)$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
-1	4	-2	7	-3
0	7	3	-1	-2

10. Use chain rule to evaluate the following derivatives:

- (a). If $x = \tan u$ and $u = t^3 - 2t \cos t + 5$, then find $\frac{dx}{dt}$
- (b). If $y = \cot^3 \sqrt{2 - 3 \sin x}$, then find $\frac{dy}{dx}$
- (c). Find $\frac{dy}{dx}$ of the functions $y = \sqrt{x^4 + \cos 3x}$ and $y = x^3 \sec \frac{1}{x}$
- (d). Find $\frac{dw}{dt}$, if $w = \sin 2x$ and $x = \sqrt{1 + \cos t^2}$
- (e). Find $\frac{dw}{dt}$, where $w = x^2$, $x = \sin y$, $y = \sqrt{t}$
- (f). Find $\frac{dy}{dx}$ for $y = \frac{(\ln(\tan x))^2}{x^3}$

Assignment-02

11. Find the equation of tangent line to the curve $f(x) = 2\tan(4x^2)$ at $x = \sqrt{\pi}$.
12. Find the equation of tangent line to the curve $f(x) = x^2 - x \sin 5x$ at $x = \pi$.
13. Evaluate the following integrals:

(a). $\int \frac{x^2}{\sqrt{1-4x^6}} dx$

(b). $\int x \tan^{-1} 2x dx$

(c). $\int \frac{\cos(\sqrt{2x-3})}{\sqrt{2x-3}} dx$

(d). $\int \frac{x^2}{\sqrt{8+x^3}} dx$

(e). $\int y\sqrt{y-1} dy$

(f). $\int \frac{dx}{\sqrt{x} e^{\sqrt{x}}}$

(g). $\int \frac{\cos 3x dx}{(1-3 \sin 3x)^3}$

(h). $\int \sqrt{2-x^2} dx$

(i). $\int x^2 \sqrt{1-x} dx$

(j). $\int \frac{\cos \frac{1}{x}}{5x^2} dx$

14. Graph the function over the specified interval. Then use simple area formula from geometry to find the area function $A(x)$ that gives the area between the graph of the specified function f and the interval $[a, x]$. Confirm that $A'(x) = f(x)$. Also, verify your answer by direct integration.

(a). $f(x) = 6 - x$; $[a, x] = [-2, x]$

(b). $f(x) = 2x + 4$; $[a, x] = \left[\frac{1}{2}, x\right]$

(c). $f(x) = 2x - 3$; $[a, x] = [2, x]$

15. Use the method of anti-derivative to find the area under the curve $y = 4(x^3 - 4x)$ within the interval $[2, 4]$.
16. Sketch the region whose area is represented by the definite integral, and evaluate the integral using an appropriate formula from geometry.

a. $\int_{-2}^2 \sqrt{4-x^2} dx$

b. $\int_1^5 |2x-5| dx$

17. Evaluate the integral $\int_{-3}^3 f(x)dx$ and $\int_{-7}^5 f(x)dx$ for $f(x) = \begin{cases} x+5, & x < 0 \\ |x-5|, & x \geq 0 \end{cases}$.

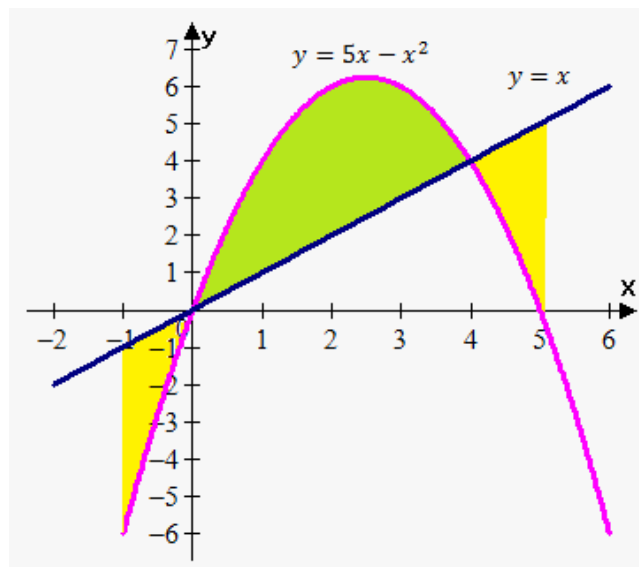
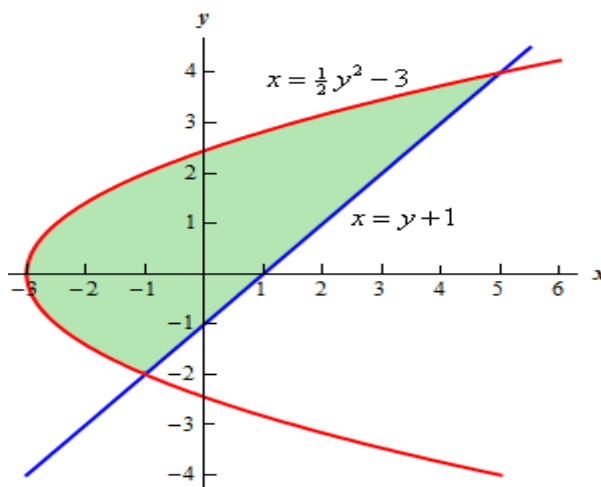
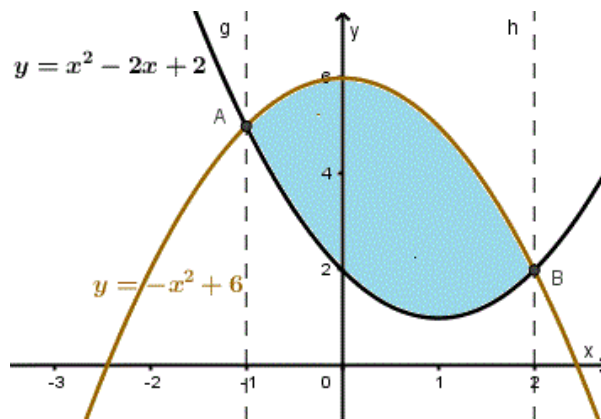
18. Evaluate the integral $\int_{-3}^{10} f(x)dx$ geometrically for $f(x) = \begin{cases} 3-x; & x \leq 5 \\ 2; & x > 5 \end{cases}$.

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19. Sketch the area represented by $f(x) = \begin{cases} 2x, & 0 \leq x \leq 2 \\ 4, & 2 < x \leq 4 \end{cases}$ for $\int_0^4 f(x) dx$. Evaluate the integral using Fundamental theorem of Calculus.
20. Graph the function $f(x) = \begin{cases} 2; x < 0 \\ x + 2; x > 0 \end{cases}$ over the interval $[-2, 4]$. Then use simple area formula from geometry to find the area within the given interval. Also, verify your answer by direct integration.
21. Use appropriate formula of geometry to evaluate the followings.
- $\int_{-6}^6 (|x - 3| + \sqrt{36 - x^2}) dx$
 - $\int_{-1}^1 (2x + \sqrt{1 - x^2} + 3) dx$
 - $\int_{-3}^3 (2 - \sqrt{9 - x^2}) dx$
22. Evaluate the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin x dx$ by the method of geometry.
23. Sketch the region whose area is represented by the $\int_{-1}^2 |2x - 1| dx$ and evaluate the integral using an appropriate formula from geometry. Also verify your result by usual integration.
24. Graph the function $f(x) = \begin{cases} 2; x < 0 \\ x + 2; x > 0 \end{cases}$ over the interval $[-2, 4]$. Then use simple area formula from geometry to find the area. Also, verify your answer by direct integration.
25. Sketch and find the total area under the curve $y = 1 - e^x$ within the interval $[-1, 1]$.
26. Sketch and find the total area under the curve $y = 1 - x^3$ within the interval $[-1, 3]$.
27. Sketch and find the total area under the curve $y = 1 - x^2$ within the interval $[-1, 2]$.
28. Find the area of the region enclosed by the following curves in the ways: **(i)** integrating with respect to x **(ii)** integrating with respect to y .
- $y^2 = 4x$ and $y = 2x - 4$
 - $x = 4y$ and $x = y^2$
 - $y = 3x^2$ and $y = 9x$
 - $y = -2x$ and $y = \frac{1}{2}x^2$
 - $x^2 = 4y, x = y - 3$
 - $x = \frac{y}{2}$ and $y = 3 - x^2$
29. Sketch the area represented by $f(x) = \begin{cases} \sqrt{x}, & 0 \leq x \leq 1 \\ 1/x^2, & 1 < x \leq 4 \end{cases}$ for $\int_0^4 f(x) dx$. Evaluate the integral using Fundamental theorem of Calculus.

Assignment-02

30. Find the area of the shaded area in the following figures.



Assignment-02

